

Back-Propagation Neural Network Versus Logistic Regression in Heart Disease Classification



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Abstract Globally, cardiovascular (heart) diseases are the major cause of death. About 80% of deaths are reported in developing countries. Looking at the trend and lifestyle, one can predict that by 2030 around 23.6 million people may die due to heart disease (mainly from heart attacks and strokes). Each and every healthcare unit generates enormous heart disease data which unfortunately are not “mined” to discover pattern and knowledge for effective decision making. Practical knowledge by domain experts plays vital role. However, there is a need for effective analysis tools to discover unknown relationships and trends in data. Objective of this paper is to assess the accuracy of classification model for the prediction of heart disease for Cleveland dataset. A comparative study of parametric and nonparametric approach in classifying heart disease is presented. Two classification models, back-propagation neural network (BPNN) and logistic regression (LR), are used for the study. The developed classification model will assist domain experts to take effective diagnostic decision. 10-fold cross validation method is used to measure the unbiased estimate of these classification models.

Keywords Heart disease • Prediction • Artificial neural network
BPNN • Accuracy

1 Introduction

In US, approximately 8 lakh death is reported due to cardiovascular disease (CVD) in 2017. On a global level, this CVD accounts for 31% (17.7 million) of all deaths. CVD, cardiomyopathy, and coronary heart disease are various forms of heart diseases which may result due to various factors such as blood pressure, obesity, smoking. Accurate diagnosis of heart disease at affordable price is a great

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© Springer Nature Singapore Pte Ltd. 2019
J. K. Mandal et al. (eds.), *Advanced Computing and Communication Technologies*,
Advances in Intelligent Systems and Computing 702,
https://doi.org/10.1007/978-981-13-0680-8_13

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challenge [1]. High-quality clinical decisions lead to efficient diagnosis, while poor-quality decisions lead to improper diagnosis of the patient. The need of the hour is high-quality clinical decisions at low cost. This can be achieved by applying advanced data mining techniques which helps in discovering hidden patterns in healthcare datasets [2, 3].

The artificial neural network (ANN) and LR, the machine learning algorithms are used in healthcare industry to minimize the diagnosis cost [4, 5]. Both parametric and nonparametric models are used to solve classification problems. Classification accuracy in ANN can be enhanced by rearranging the weights of neurons [6]. ANN and LR with decision support play a very important role in improving the quality of service in health care or in medical centers. Thus, ANN model-based diagnoses have significantly helped health care by providing effective diagnosis [7].

1.1 Artificial Neural Network Architecture (ANN)

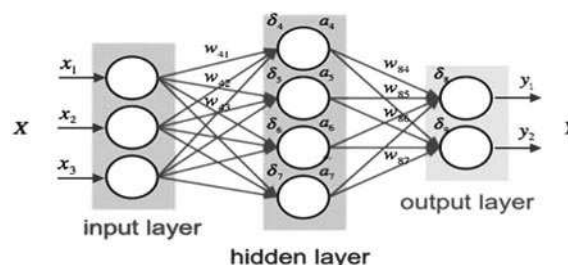
Artificial neural network can be divided into three layers, namely input layer, hidden layer, and output layer [8]. Figure 1 shows general structure of ANN [9].

- Input layer: This layer receives data from the external environment. These inputs are usually normalized which results in better numerical precision for the mathematical operations performed by the network.
- Hidden layers: These layers are composed of neurons which are responsible for extracting patterns associated with the data that is fed to the neural network.
- Output layer: The results of the processing performed by the neurons in the previous layers are produced by these neurons.

1.1.1 Number of Hidden Neurons

Selection of number of hidden neurons is very important for getting good results. Inappropriate number of hidden neuron causes either overfitting or underfitting. If the number of neurons is less as compared to the complexity of the problem data,

Fig. 1 ANN architecture.
Courtesy A Medium Corporation



then “Under fitting” may occur. If the number of neurons is more for the complexity of problem, then “Over fitting” may occur. Till now, there is no exact formula for computing the number of hidden neurons. Most of the times, it is trial and error method.

However, there are several rule-of-thumbs for choosing the number of hidden neurons.

- The number of hidden layer neurons is $2/3$ of the size of the input layer [10]
- The number of hidden layer neurons should be less than twice of the number of neurons in input layer [3]
- The size of the hidden layer neurons is between the input layer size and the output layer size [11].

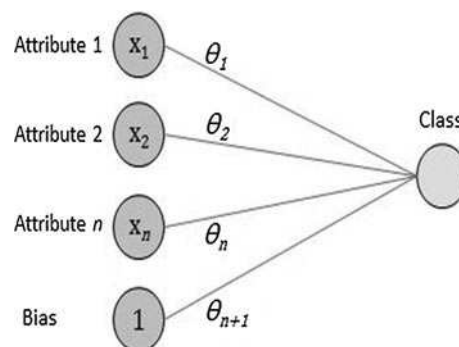
1.1.2 Back-Propagation Algorithm

Back-propagation algorithm is a popular supervised learning model [11]. The BPNN adopts the principle of multi-layer feed-forward network which models a function by performing error calculation of each neuron [12]. BPNN keeps on updating its weight after processing the group of data until error values are within the threshold [10, 13].

1.2 Logistic Regression

LR is a parametric model where the categorical variables are dependent [14]. This parametric model is used for classifying categorical values, whose values depend on the success of probability. Figure 2 depicts the LR model. It involves fitting an equation of the form:

Fig. 2 Logistic regression model



$$i = 1 / \left(1 + e^{((x0 + x1) * j)} \right) \quad (1)$$

i = input values, j = output values, $x0$ = bias term, $x1$ = coefficient for single input value (i).

2 Related Work

Vikas et al. (2013) proposed data mining approach to detect heart disease using Naive Bayes, decision tree, and bagging algorithm. In this paper, a systematic comparison is carried out to between bagging and non-bagging algorithms and accuracy was found to be 85.03% on bagging algorithm [15].

Kirmanani [16] proposed a prediction model using multi-layer perceptron algorithm. Input dataset with 13 attributes and 303 records was made as input to the algorithm. Out of which, 245 records were accurately diagnosed with accuracy of 80.85%.

Abdar et al. [17] proposed a comparative study of various data mining approaches for prediction of heart disease. Input datasets with 13 attributes and 270 records were made as input to the several algorithms like C5.0, support vector machine (SVM), k-nearest neighbor (k-NN), and neural networks. Decision tree algorithm-C5.0 model, found to produce highest accuracy of 93.02%.

Gandhi and Singh [14] proposed a comparative study on classification model in predicting heart disease using decision tree, neural network, and Naive Bayes. Through the comparative study, a better hybrid model could be built in predicting heart disease.

Sayed and Halkarnikar [18] proposed a genetic neural-based algorithm for prediction of heart disease. Input dataset with more than 11 attributes was made as input to the algorithm. The model was implemented in MATLAB environment, and accuracy was found to be 89% [16, 18].

Sayad et al. [19] proposed a neural network-based approach for diagnosis of heart disease. In this approach, a multi-layer perceptron with BPNN model is built to train the input datasets with 13 attributes and 70 records. The accuracy of the model was found to be 94%. This paper is considered as a benchmark, and results are compared with proposed method [19].

Giraddi et al. [20] proposed a system based on BPNN for the detection of diabetic retinopathy (DR). DR is a complication in which vision of a person gets affected by long-term diabetes. The author explored various architectures to find best model for the detection of diabetic retinopathy.

Research gap noticed through the survey is: It is observed that various machine learning methods are employed for identifying accurate classification model for predicting heart disease. Thirteen attributes have become de facto dataset for experimentation. There is a need for comparing parametric and nonparametric approach in classifying toward predicting heart disease.

3 Data Description

Cleveland heart disease dataset has been used for the study [21]. Database consists of 303 instances with 13 attributes each, out of which 270 instances are used for the study. The database is prepared by V.A. medical center, long beach, and Cleveland clinic foundation. Description of attributes used for classification is presented in Table 1. Dataset is used for academic research with due permission request to concerned authority.

Table 1 Dataset description

Attribute number	Attributes	Description	Range
1	Age	Age in year	Continuous
2	Sex	(1 = male, 0 = female)	Discrete
3	Cp	Value 1: typical angina Value 2: atypical angina Value 3: non-angina pain Value 4: asymptomatic	Discrete
4	Trestbps	Resting blood pressure (in mm Hg on admission to hospital)	Continuous
5	Chol	Serum cholesterol in mg/dl	Continuous
6	Fbs	(Fasting blood sugar > 120 mg/dl) (1 = true, 0 = false)	Discrete
7	Restecg	Resting electrocardiographic results Value 0: normal Value 1: having ST-T wave abnormality (T wave inversions and/or ST Elevation or depression of > 0.05 mV) Value 2: showing probable or definite left ventricular hypertrophy by Estes criteria	Discrete
8	Thalach	Maximum heart rate achieved	Continuous
9	Exang	Exercise induced angina (1 = yes, 0 = no)	Discrete
10	Oldspeak	ST depression induced by exercise relative to rest	Continuous
11	Slope	The slope of the peak exercise ST segment Value 1: up sloping Value 2: flat Value 3: down sloping	Discrete
12	Ca	Number of major vessels (0–3) colored by fluoroscopy	Continuous
13	Thal	Normal fixed defect, reversible defect	Discrete

4 Methodology

The implementation of models is carried out using Anaconda Python language.

- Normalization: The dataset has been normalized to have a mean of zero and variance 1.
- Hidden layers and hidden neurons: The number of neurons is decided based on various thumbs-of-rules given in Sect. 1.
- Training Parameters: The network has been trained using gradient descent algorithm in batch mode (trained). The number of epochs is set to 200. Learning rate is set to 0.05.

4.1 Performance Evaluation Parameters

Performance of proposed classification models is assessed by well-known parameters such as accuracy, receiver operating curve (ROC), and confusion matrix.

4.2 Hypotheses Testing

In this section, different hypothesis testing such as *T*-test, *Z*-test, *F*-test, and chi-square test which are used for testing of null hypothesis is given below. The results of these testing are used to optimize the input parameters.

Z-test

$$Z_{\text{data}} = \frac{p1 - p2}{\sqrt{P_{\text{pooled}} * (1 - P_{\text{pooled}}) \left(\frac{1}{n^1} + \frac{1}{n^2} \right)}} \quad (2)$$

where

$$P_{\text{pooled}} = \frac{x1 + x2}{n1 + n2}$$

T-test

$$t_{\text{data}} = \frac{\bar{x}1 - \bar{x}2}{\sqrt{\frac{s1^2}{n1} + \frac{s2^2}{n2}}} \quad (3)$$

F -test

$$\text{MSTR} = \frac{\sum n1(x1 - \bar{x})^2}{k - 1}, \text{MSE} = \frac{\sum (n1 - 1)S1^2}{n_t - k} F_{\text{data}} = \frac{\text{MSTR}}{\text{MSE}} \quad (4)$$

(annova test)

Chi-square-test

$$\text{Chi}_{\text{data}}^2 = \sum \frac{(O - E)^2}{E} \quad (5)$$

- (a) x_1, x_2 —elements belonging to different group b. O —observed frequency c. E —expected frequency

5 Results and Discussion

5.1 Back-Propagation Neural Network (BPNN) Versus Linear Regression

The proposed heart disease classification is carried out with different architecture of ANN using back-propagation algorithm. The results of 10-fold cross validation for these architectures are given in Table 2. Table 3 shows the confusion matrices obtained for 27 samples. Figure 3 shows the classification accuracies obtained. Figure 4 shows the ROC curve for BPNN and that of Fig. 5 is ROC for linear regression.

Table 2 Percentage of MSE accuracy with BPNN architecture

Architecture	MSE	Accuracy (%)
13-6-6-1	0.2962	72.72
13-6-6-6-1	0.1851	85.07
13-6-6-6-6-1	0.2222	79.25
13-9-9-9-1	0.1851	78.51
13-12-12-1	0.2222	78.88

Table 3 Confusion matrices for various architectures

Architecture	PT	FP	TN	FN	SN (%)	SP (%)
13-6-6-1	0.3407	0.437	0.118	0.103	76.66	70.66
13-6-6-6-1	0.3444	0.4518	0.103	0.1	77.5	71.56
13-6-6-6-6-1	0.33	0.455	0.1	0.107	75.83	69.21
13-9-9-9-1	0.329	0.459	0.962	0.114	76.66	71.37
13-12-12-1	0.318	0.466	0.88	0.318	74.16	70.70

Fig. 3 Classification accuracies with various architectures

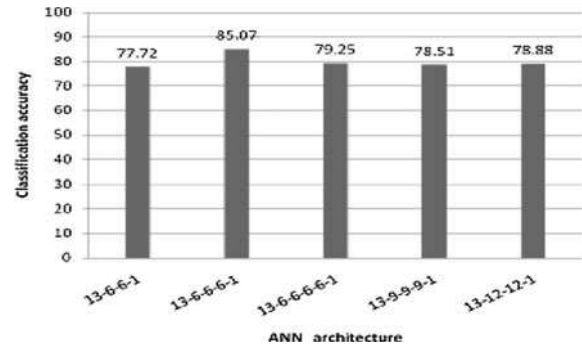


Fig. 4 ROC for BPNN

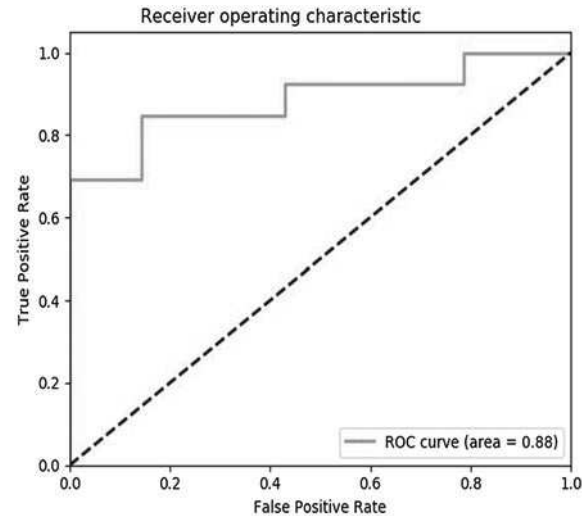
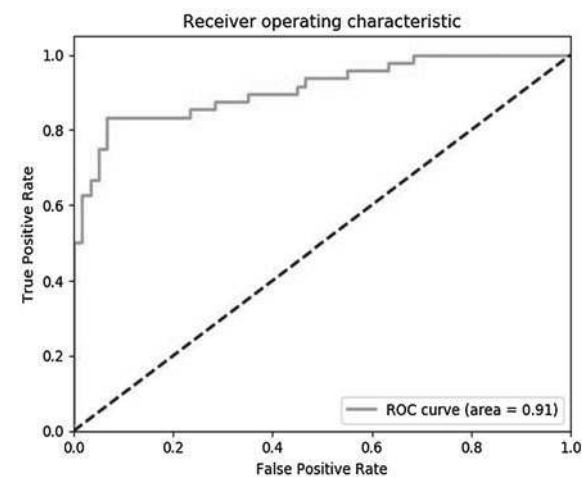


Fig. 5 ROC for linear regression



5.2 Hypothesis Testing and Analysis

It is very important to study each and every attribute and its influence on the classification model. Hence, following null hypothesis testing is defined and validated using P -test.

Case 1: P -test is conducted to test to check is there any significant difference in mean cholesterol of population significantly vary with respect to heart disease.

Inference: The observed p -value > 0.025 (95% CI); hence, it fails to reject the null hypothesis. There is no significant difference in cholesterol, with respect to heart disease. This indicates cholesterol is not one of the risk factors related to heart disease.

Case 2: Is there any significant difference in mean cholesterol of population significantly vary with respect to heart disease.

Inference: The observed p -value < 0.025 (95% CI); hence, we have evidence to reject the null hypothesis. There is a significant difference in blood pressure, with respect to heart disease. This indicates that prolonged high BP is one of the risk factors to get into a heart disease.

Case 3: Is there any significant difference in mean age of population significantly vary with respect to heart disease.

Inference 3: The observed p -value < 0.025 (95% CI); hence, we have evidence to reject the null hypothesis. There is a significant difference in age, with respect to heart disease. This indicates that age is one of the risk factors to get a heart disease.

Case 4: Is there any significant difference in mean age of population significantly vary with respect to heart disease.

Inference: The observed p -value < 0.025 (95% CI); hence, it fails reject the null hypothesis. There is a significant difference in heart rate, with respect to heart disease. This indicates that heart rate is one of the risk factors to get into a heart disease

Table 4 presents the p -value obtained for all the above four cases. In order to study the effect of more than two input attributes and their impact on the heart disease classification, Z -test, F -test, and chi-square tests are carried out.

Case 5: Whether exercise-doing people have a less risk of heart disease.

Inference: The observed p -value < 0.025 (95% CI); hence, it rejects the null hypothesis. There is a significant difference in exercise, with respect to heart disease. Hence, exercise-doing people have less risk of heart disease when compared to non-exercise people.

Table 4 P -test results

Number of case	Heart disease	p_value
Case 1	Mean cholesterol	0.05273
Case 2	Mean BP	0.01056
Case 3	Mean age	0.00044
Case 4	Mean heart rate	7.1103e-13

Table 5 Z-test results

Number of cases	Heart disease	Z_value	P-value
Case 5	Exercise	-10.365	0.0
Case 6	Gender	-11.354	0.0

Case 6: Whether the proportion of heart disease varies with respect to gender (male/female).

Inference: The observed p -value < 0.025 (95% CI); hence, it rejects the null hypothesis. There is a significant difference in gender, with respect to heart disease. Hence, based on the gender, there is a significant difference with respect to heart disease. Table 5 presents the z -value and p -value obtained for both the case.

Case 7: To check whether there is any significant difference with mean cholesterol with respect to vessels.

Inference: The observed p -value > 0.025 (95% CI); hence, it fails to reject the null hypothesis. That is there is no significant difference in mean cholesterol level with respect to vessel blockage.

Case 8: To check whether there is any significant difference of mean age with respect to vessels.

Inference: The observed p -value < 0.025 (95% CI); hence, it rejects the null hypothesis. That is there is a significant difference in age with respect to vessel blockage.

Table 6 presents the calculated p -value and F -value with respect to MSE and MSTR

Case 9: To check whether there is any significant difference between the genders with respect to vessels blockage.

Inference: The observed p -value > 0.025 (95% CI); hence, it fails to reject the null hypothesis. There is no significant difference in gender with respect to vessels. Which indicates gender is not one of the risk factors related to vessels.

Table 7 presents the calculated chi-square test and its corresponding p -value.

Results of proposed BPNN-based heart disease classification are compared with the recent literature [16], where multi-layer perceptron neural network is employed for the same classification and is presented in Table 8.

It is worth noting that accuracy of BPNN-based classification model is approximately 5% more as compared to the recent literature [16].

Table 6 F-test results

Number of case	Vessels	MSTR	MSE	$F = \text{MSTR}/\text{MSE}$	P-value
Case 7	Cholesterol	4162.7	2654.6	1.5681	0.1974
Case 8	Age	1057.0	71.98	14.683	7e-09

Table 7 Chi-square results

Number of cases	Chi-square test	P-value
Case 9	6.4626	0.039506

Table 8 Results of proposed methods with other related works

Methodology	Datasets and records	Accuracy (%)	Author
Multi-layer perceptron BPNN	13 attributes 303 records	80.85	[1, 16]
BPNN	13 attributes 270 records	85.07	Proposed
LR	13 attributes 270 records	92.58	Proposed

6 Conclusions and Future Scope

Two objectives of this paper are identifying the best classification model among parametric and nonparametric for effective heart disease prediction and optimizing the number input attributes for classification model which were achieved by various systematic and engineering approaches. Experiments are conducted to find the best classifier for predicting the existence of heart disease. Input dataset having 13 fields and 270 records collected from Cleveland heart disease dataset is employed to validate the system. Among 13 fields, only 11 fields are most influencing and contributing parameters. This is achieved by testing various null hypotheses and cross-correlation matrix. The output variable is categorical in nature and the parametric based algorithms outputs are found to be more effective. Accuracy of 85.074 and 92.58% is recorded for BNN (nonparametric) and LR (parametric) models, respectively. With 10-fold cross validation and ROC analysis, area of 0.88 and 0.91 is recorded for BNN and LR, respectively. The empirical results show that with parametric model-LR, we can test effectively any new case of heart disease patient with optimal 11 parameters. At any moment of time, the model proposed cannot be replacement for clinical experts. It only complements the decision of clinician for taking better diagnostic decisions.

The futuristic scope of this project is to collect the dataset from local regional healthcare unit, along with additional input attributes such as smoking/alcoholic habits, food habit, and genetic conditions. Apply well-known parametric classification model and even designing hybrid model for effective heart disease prediction.

Acknowledgements Authors acknowledge “UCI Machine Learning Repository” for heart disease dataset. This work is partially supported by KLE Tech University under “Capacity Building Projects” grants. Authors acknowledge KLE Society and KLE Tech University, Hubli, for providing funding and support.

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